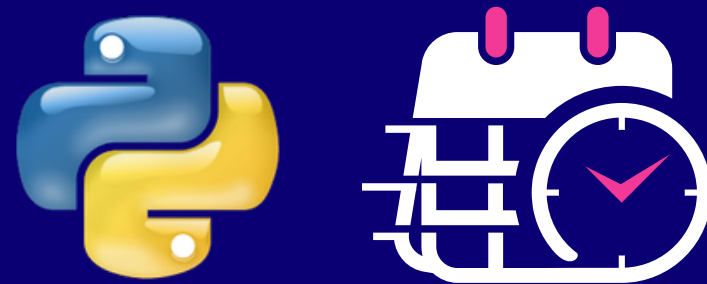


# DATA PIPELINE DOCUMENTATION: LEGACY TRANSACTION ETL

Membersihkan dan menstandarisasi data transaksi kotor dari sistem lama (legacy) ke format siap pakai (Warehouse-Ready)



Python3 & Datetime Module

# Data Profiling

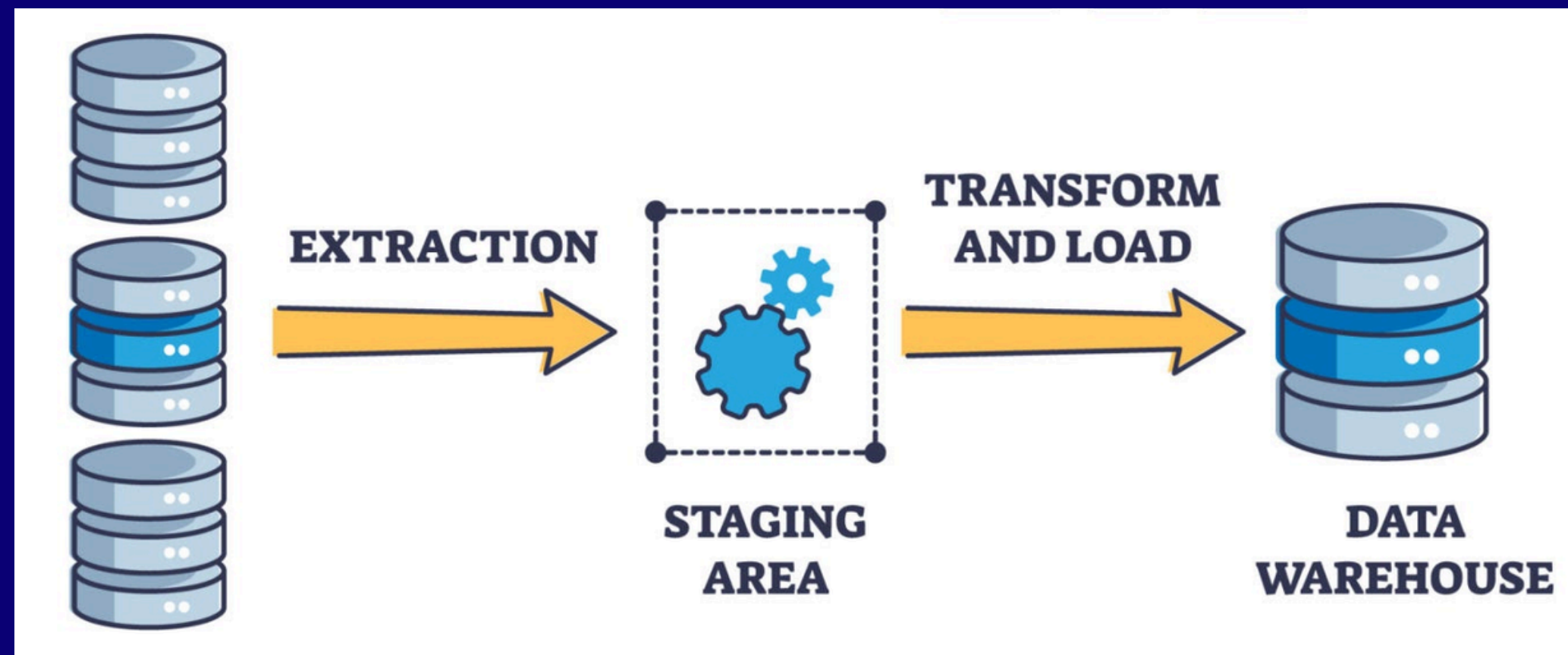
Before

|   | tx_id   | date       | product             | price    | qty |
|---|---------|------------|---------------------|----------|-----|
| 0 | TXN-001 | 2023-10-01 | Laptop Gaming       | 15000000 | 1   |
| 1 | TXN-002 | 2023-10-01 | Mouse Wireless      | 150000   | 2   |
| 2 | TXN-003 | 02/10/2023 | Monitor 24 inch     | 2000000  | 1   |
| 3 | TXN-004 | 2023-10-02 | HDMI Cable          | None     | 5   |
| 4 | TXN-005 | 2023-10-03 | KEYBOARD MECHANICAL | 500000   | 1   |

After

|   | transaction_id | date       | product_name        | total_sales |
|---|----------------|------------|---------------------|-------------|
| 0 | TXN-001        | 2023-10-01 | Laptop Gaming       | 15000000.0  |
| 1 | TXN-002        | 2023-10-01 | Mouse Wireless      | 300000.0    |
| 2 | TXN-003        | 2023-10-02 | Monitor 24 Inch     | 2000000.0   |
| 3 | TXN-005        | 2023-10-03 | Keyboard Mechanical | 500000.0    |

# Logika ETL



# Logika ETL

## 1. Extraction (dummy)

```
raw_data = [  
    {"tx_id": "TXN-001", "date": "2023-10-01", "product": "Apple"},  
    {"tx_id": "TXN-002", "date": "2023-10-01", "product": "Banana"},  
    {"tx_id": "TXN-003", "date": "02/10/2023", "product": "Mango"},  
    {"tx_id": "TXN-004", "date": "2023-10-02", "product": "Orange"}  
]
```

## 2. Transformation

Helper Function String Cleaning:

```
def clean_text(product_name):  
    '''pastikan yg masuk ke fungsi ini ada'''  
    if not isinstance(product_name, str):  
        return "Unkown"  
    return product_name.strip().title()
```

# Logika ETL

Helper Function Datetime Standardization:

```
def standardize_date(date_str):  
    '''parsing objek String jadi Datetime'''  
    try:  
        #mencoba parsing format YYYY-MM-DD  
        dt = datetime.strptime(date_str, "%Y-%m-%d")  
        return dt.strftime("%Y-%m-%d")  
    except ValueError: #kalo error / td  
        #mencoba parsing format DD/MM/YYYY  
        try:  
            dt = datetime.strptime(date_str, "%d/%m/%Y")  
            return dt.strftime("%Y-%m-%d")  
        except:  
            return None  
    ...
```



# Logika ETL

Extraction & Validation:

```
for row in raw_data:
    # A. Data Extraction & Validation (Variables & If-Else)
    price = row.get("price")
    qty = row.get("qty")

    # F (variable) price: Any price kosong (Data Quality)
    if price is None:
        print(f"Skipping corrupt data: {row['tx_id']}")
        continue #lanjut ke data berikutnya (kalo gk corr
```



# Logika ETL

Type Casting (string to int/float) & Calculation:

```
try:
    # Type Casting: String ke Float/Int
    price_float = float(price)
    qty_int = int(qty)

    #String Manipulation
    cleaned_product = clean_text(row["product"])

    #Date Handling
    cleaned_date = standardize_date(row["date"])

    if cleaned_date is None:
        print(f"Invalid date for {row['tx_id']}")
        continue

    #Calculation
    total_sales = price_float * qty_int
```

# Logika ETL

## 3.Loading

```
transformed_row = {  
    "transaction_id": row["tx_id"],  
    "date": cleaned_date,  
    "product_name": cleaned_product,  
    "total_sales": total_sales  
}  
  
clean_data.append(transformed_row)  
  
#agg  
total_revenue += total_sales
```

```
except ValueError as e:  
    print(f"Error processing row {row["tx_id"]}: {e}")
```



# Logika ETL

Output:

```
C---Memulai Proses ETL---  
Skipping corrupt data: TXN-004  
  
--- Data bersih (ready for warehouse) ---  
{'transaction_id': 'TXN-001', 'date': '2023-10-01', 'product_name': 'Laptop Gaming', 'total_sales': 150000  
00.0}  
{'transaction_id': 'TXN-002', 'date': '2023-10-01', 'product_name': 'Mouse Wireless', 'total_sales': 30000  
0.0}  
{'transaction_id': 'TXN-003', 'date': '2023-10-02', 'product_name': 'Monitor 24 Inch', 'total_sales': 2000  
000.0}  
{'transaction_id': 'TXN-005', 'date': '2023-10-03', 'product_name': 'Keyboard Mechanical', 'total_sales':  
500000.0}  
  
--- Total Revenue: 17,800,000.00
```

# Kesimpulan

**Key Metrics:** Total Revenue yang berhasil dihitung (Rp 17,800,000.00).

```
--- Total Revenue: 17,800,000.00
```

**Data Integrity:** "1 record dihapus karena tidak terdapat nilai "price" (Data Quality Guard)."

```
Skipping corrupt data: TXN-004
```

# THANK YOU

